



Crusader Colony Sim



Core vision: CK3-scale people and politics + RimWorld-style pawns and bodies + live RTS/open-world simulation + shonen-grade combat expressiveness + browser-native execution.

Crusader Colony Sim is a multiscale agent simulation where bodies, societies, combat techniques, and history emerge from composable mathematical systems rather than scripted content.

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0. World Laws

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0. World Laws

The foundational rules shared by every entity and system.

- **Space and geometry** — position, distance, shape, collision, and spatial relationships
 - **Time and simulation ticks** — ordering, duration, scheduling, and rates of change
 - **Matter, terrain, and structures** — physical composition and the world's persistent form
 - **Resources and energy** — storage, transfer, consumption, and regeneration
 - **Information propagation** — how signals, knowledge, and effects travel
 - **Visibility and uncertainty** — what can be perceived, inferred, hidden, or mistaken
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1. Pawn Model

A pawn should eventually be much richer than `position + hp`.

$$P = (B, K, M, C, S, R, I, G)$$

Symbol	Component	Governing question
B	Body topology	What are you made of?
K	Kinematics	How can you move?
M	Material / physical state	What condition are you in?
C	Cognition	What do you perceive, know, and decide?
S	Skills	What have you learned to do?
R	Relationships	Who matters to you, and how?

Symbol	Component	Governing question
I	Inventory / equipment	What do you possess and wield?
G	Goals	What are you trying to achieve?

Each component is its own graph. Together, these graphs govern the pawn's capabilities, constraints, decisions, and observable outcomes.



That abstraction is what lets a sword technique, telekinetic shove, cursed-energy blast, broken arm, prosthetic leg, cape, shield, giant hammer, or bizarre anime stance all interact with the **same underlying pawn system**.

2. Foundational Body Graphs

G₁ — Anatomical Graph

- **Nodes:** body parts, organs, tissues, prosthetics
- **Edges:** attachment, containment, coverage, support
- **State:** HP, integrity, capacity, impairment

| **Governs:** what you can do.

G₂ — Kinematic Graph

- **Nodes:** joints and articulated links
- **Edges:** degrees of freedom, movement limits, mechanical coupling
- **State:** mass, inertia, pose, velocity

| **Governs:** how you move.

G₃ — Conduit Graph

- **Nodes:** channels, reservoirs, sources, and sinks
- **Edges:** flow paths and transfer relationships
- **State:** capacity, throughput, saturation, integrity

| **Governs:** what impossible things you can do — and what they cost.

G₃ — Conduit Graph

3. Combat as State Transformation

Combat is modeled as **state transformations over articulated bodies and the surrounding world**. An attack becomes:

$$A : (P_a, P_d, W, t) \rightarrow \Delta(P_a, P_d, W)$$

with components:

$$A = (\text{trajectory, force, energy, target, timing, constraints, effects})$$

Technique routing

A technique doesn't set values directly — it declares a **routing**: a required flow from source to effector across G_3 . Solving for actual per-channel throughput is a flow problem on a weighted graph, i.e. a Laplacian solve. Kirchhoff's laws for k_i .

Without special-casing anything, you get:

- **Gear Second** — deliberately over-drive the cardiovascular trunk. $\tau \gg C$. Enormous output, **s** spikes, **I** bleeds, you pay later. A parameter regime, not a scripted state.
- **Eight Inner Gates** — sequential release of capacity limiters, each unlock raising **c** while multiplying μ . The eighth gate isn't a cutscene; it's $\mu \rightarrow \infty$.
- **Gentle Fist** — a damage type that targets G_3 edges instead of G_1 parts. A whole martial school falls out of one damage channel.
- **One For All @ 20%** — output ceiling derived from limb **I** and **c**. The number *means* something.
- **Bloodline techniques** — high inherited **c** concentrated in a region (ocular, spinal, dermal). Dynastic genetics become mechanically real without a single hardcoded trait — exactly where CK3 fuses in.

4. Simulation Scales



Pawn scale

Milliseconds to seconds. Bodies, attacks, movement, wounds, perception, projectiles.



Colony scale

Minutes to months. Jobs, production, construction, logistics, families, social structures, factions.



Dynasty / world scale

Years to centuries. Inheritance, diplomacy, religion, ideology, wars, migration, technological change.

The scales close into a loop:

combat → injury → relationship → politics → war → combat

| A duel can literally change history.

5. Layer Budget

Each layer's output becomes the next layer's setpoint; only the physical layer sends signals back upward.

Layer	Rate	State	Output = next setpoint
Strategic (<i>CK3</i>)	event / 1 Hz	dynasty, claims, opinion, schemes, traits, stress	→ operational orders
Operational (<i>RTS</i>)	~10 Hz	armies, supply, sieges, flow fields, influence	→ tactical jobs
Tactical (<i>RimWorld</i>)	10–30 Hz	pawn utility/GOAP, needs, jobs, frame-data adjudication	→ move invocations + task objectives
Physical (<i>bodies</i>)	120 Hz fixed	articulated dynamics, PD tracking, G ₃ flow	→ torques, contacts; ↑ veto signal only

▼ 6. Capability Algebra

The shonen inspiration should become a **general capability algebra**, not a collection of special-case superpowers:

Ability = Source × Transformation × Carrier × Geometry × Target × Constraint

Ability	Decomposition
Sword slash	muscular energy → kinetic → blade → arc → body
Fire blast	internal energy → thermal → plasma/fire → cone → area
Gravity technique	ability resource → gravitational field → space → radial field → entities
Teleport strike	resource → spatial transform → pawn → point-to-point → self

The result is a **combat grammar** rather than an ability list.

▼ 7. Pawn Bodies as Graphs

The pawn body may become the real secret sauce. Instead of:

```
Pawn
  HP = 100
```

we use a graph:

```
Pawn
├── torso
│   ├── heart
│   ├── lungs
│   └── spine
├── left_arm
│   ├── humerus
│   ├── forearm
│   └── hand
└── ...
```

Crucially, the representation is **abstract enough that creatures don't need human anatomy**:

$$\text{Body} = (V, E)$$

where V are anatomical or functional parts and E represent structural, circulatory, neural, mechanical, or dependency relationships.

The same body machinery then serves a human, horse, dragon, robot, giant, slime, or cursed puppet.